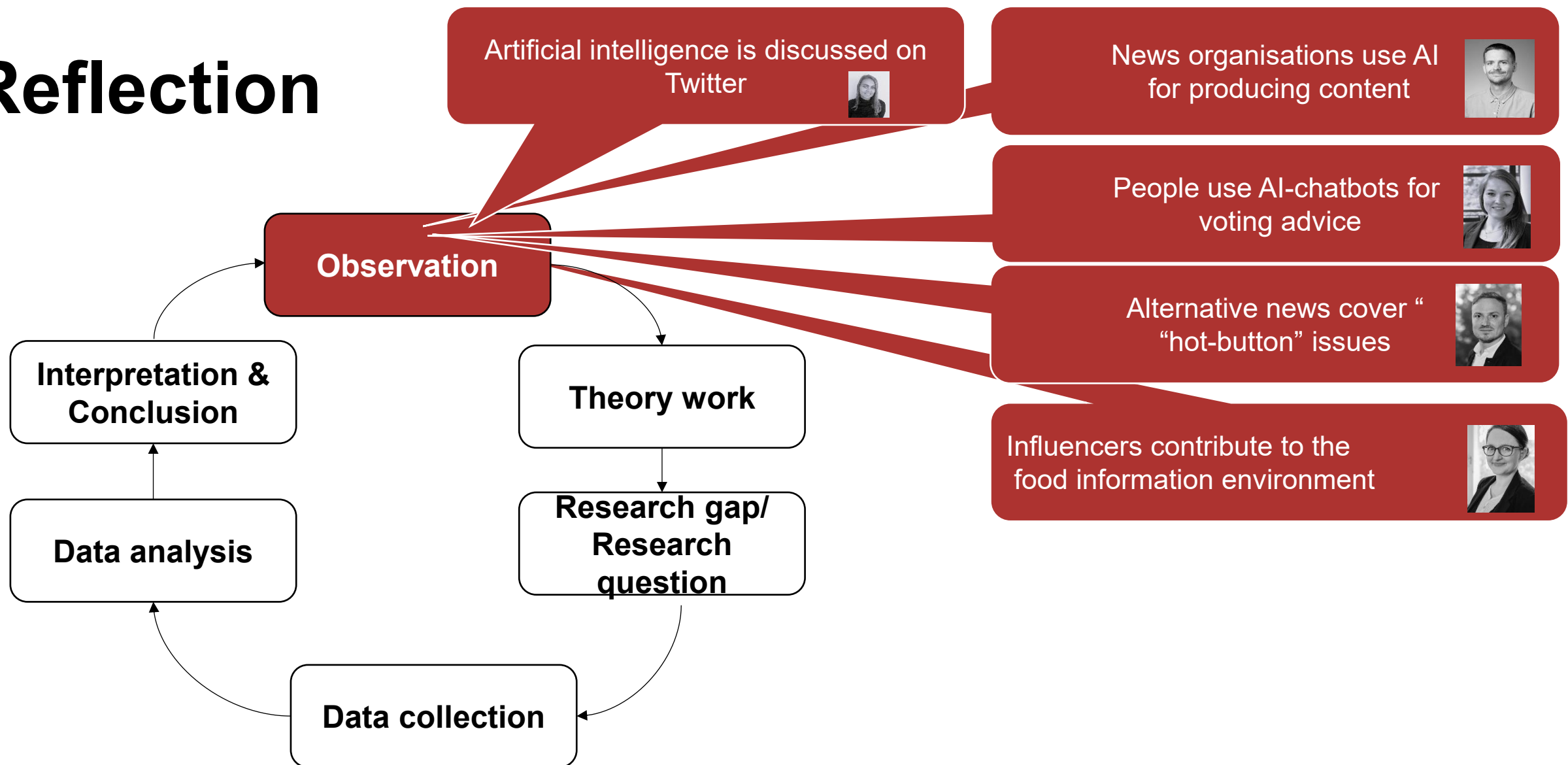


**AI 507 –
Research seminar
AI & Society**

**#RS6 Hands-on: Writing a
theory section**

Reflection



Scientific articles & reports reflect this logic

www.nature.com/scientificreports

scientific reports

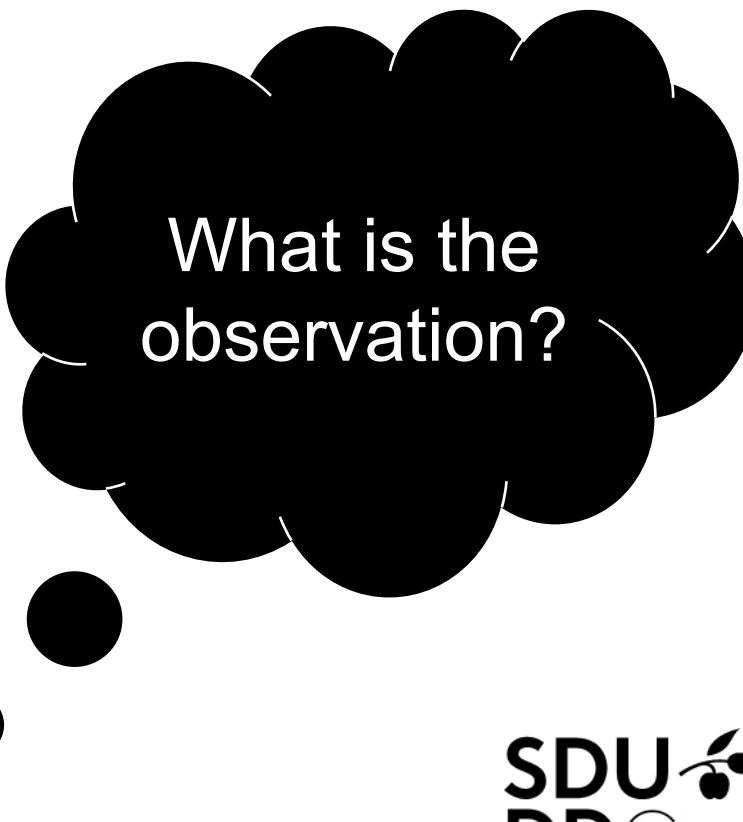
OPEN

The potential of generative AI for personalized persuasion at scale

S. C. Matz^{1,2✉}, J. D. Teeny³, S. S. Vaid⁴, H. Peters¹, G. M. Harari⁵ & M. Cerf¹

Matching the language or content of a message to the psychological profile of its recipient (known as “personalized persuasion”) is widely considered to be one of the most effective messaging strategies. We demonstrate that the rapid advances in large language models (LLMs), like ChatGPT, could accelerate this influence by making personalized persuasion scalable. Across four studies (consisting of seven sub-studies; total $N = 1788$), we show that personalized messages crafted by ChatGPT exhibit significantly more influence than non-personalized messages. This was true across different domains of persuasion (e.g., marketing of consumer products, political appeals for climate action), psychological profiles (e.g., personality traits, political ideology, moral foundations), and when only providing the LLM with a single, short prompt naming or describing the targeted psychological dimension. Thus, our findings are among the first to demonstrate the potential for LLMs to automate, and thereby scale, the use of personalized persuasion in ways that enhance its effectiveness and efficiency. We discuss the implications for researchers, practitioners, and the general public.

Financial analysts have described people’s digital behavioral data as “more valuable than oil”^{1,2}. This is, in part, because such records afford one of the most effective forms of influence: *personalized persuasion*^{3,4}. Compared to non-personalized communication, matching the content of a persuasive message (e.g., its language or visuals) to the psychological profile of its recipient enhances its effectiveness (e.g.,^{4,5}). On the one hand, such personalization offers tremendous opportunities to promote desired behaviors, including a healthy lifestyle^{6–8}, financial saving⁹, or support for environmentalism¹⁰. On the other hand, it can have a pernicious effect on societies¹¹, for example, increasing the spread of disinformation¹², manipulating political preferences^{13,14}, or promoting maladaptive consumer decision-making^{15,16}. We provide the first empirical evidence demonstrating how content generated by artificial intelligence (AI) can scale personalized persuasion by automating the creation of such messages with only limited information about the message recipient. As legislators increasingly consider whether (and how) to regulate generative AI¹⁷, our work suggests that AI-automated, personalized persuasion is poised to create an inflection point for the implementation and effectiveness of this influence tactic.



What is the observation?

Scientific articles & reports reflect this logic

www.nature.com/scientificreports

scientific reports

OPEN

The potential of generative AI for personalized persuasion at scale

S. C. Matz^{1,2✉}, J. D. Teeny³, S. S. Vaid⁴, H. Peters¹, G. M. Harari⁵ & M. Cerf¹

Matching the language or content of a message to the psychological profile of its recipient (known as “personalized persuasion”) is widely considered to be one of the most effective messaging strategies. We demonstrate that the rapid advances in large language models (LLMs), like ChatGPT, could accelerate this influence by making personalized persuasion scalable. Across four studies (consisting of seven sub-studies; total $N=1788$), we show that personalized messages crafted by ChatGPT exhibit significantly more influence than non-personalized messages. This was true across different domains of persuasion (e.g., marketing of consumer products, political appeals for climate action), psychological profiles (e.g., personality traits, political ideology, moral foundations), and when only providing the LLM with a single, short prompt naming or describing the targeted psychological dimension. Thus, our findings are among the first to demonstrate the potential for LLMs to automate, and thereby scale, the use of personalized persuasion in ways that enhance its effectiveness and efficiency. We discuss the implications for researchers, practitioners, and the general public.

Financial analysts have described people’s digital behavioral data as “more valuable than oil”^{1,2}. This is, in part, because such records afford one of the most effective forms of influence: *personalized persuasion*^{3,4}. Compared to non-personalized communication, matching the content of a persuasive message (e.g., its language or visuals) to the psychological profile of its recipient enhances its effectiveness (e.g.,^{4,5}). On the one hand, such personalization offers tremendous opportunities to promote desired behaviors, including a healthy lifestyle^{6–8}, financial saving⁹, or support for environmentalism¹⁰. On the other hand, it can have a pernicious effect on societies¹¹, for example, increasing the spread of disinformation¹², manipulating political preferences^{13,14}, or promoting maladaptive consumer decision-making^{15,16}. We provide the first empirical evidence demonstrating how content generated by artificial intelligence (AI) can scale personalized persuasion by automating the creation of such messages with only limited information about the message recipient. As legislators increasingly consider whether (and how) to regulate generative AI¹⁷, our work suggests that AI-automated, personalized persuasion is poised to create an inflection point for the implementation and effectiveness of this influence tactic.

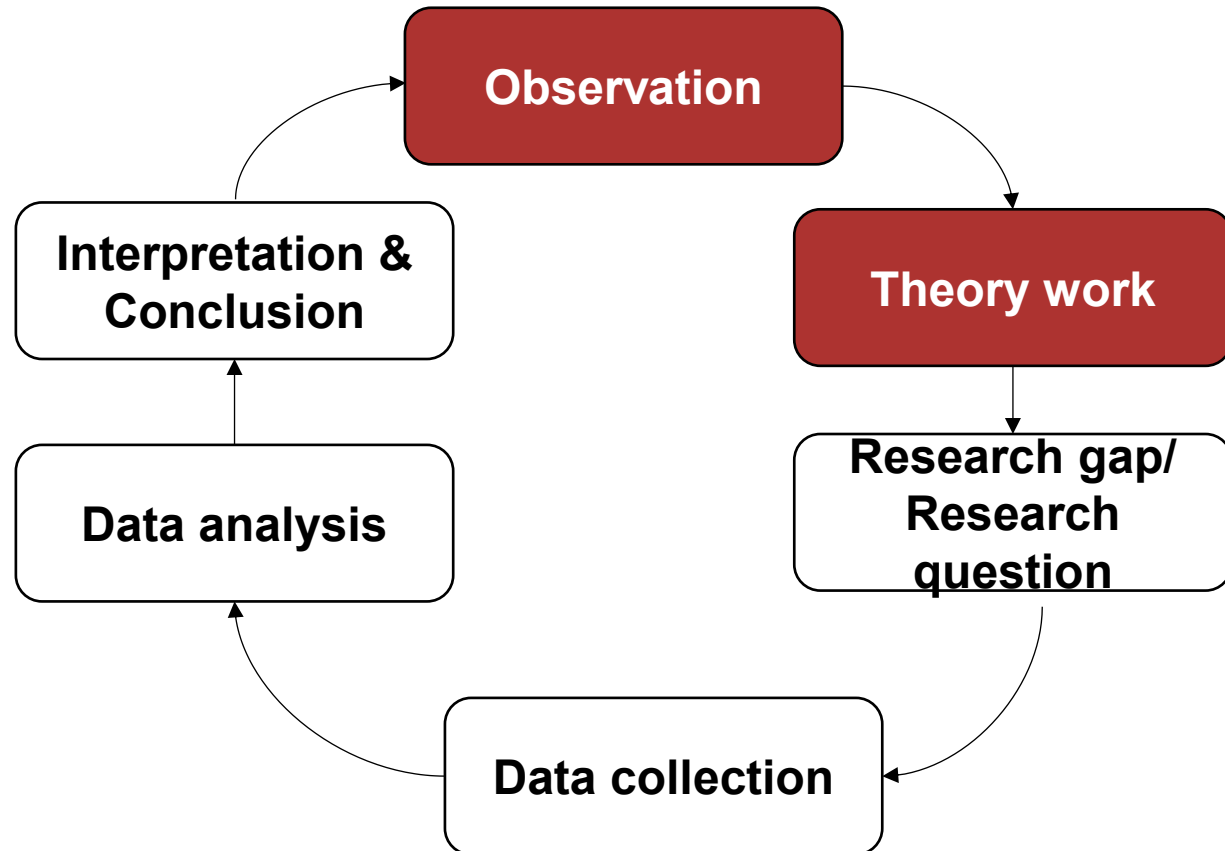
Every compelling CHI paper begins with a problem that leads to a burning research question.

This question serves as your direction. It guides your research journey.

First, define the problem you’re addressing. Why does it matter? Who is affected? What are the current limitations? Authors anchor their work in past findings and theories to showcase that the problem does not stand isolated

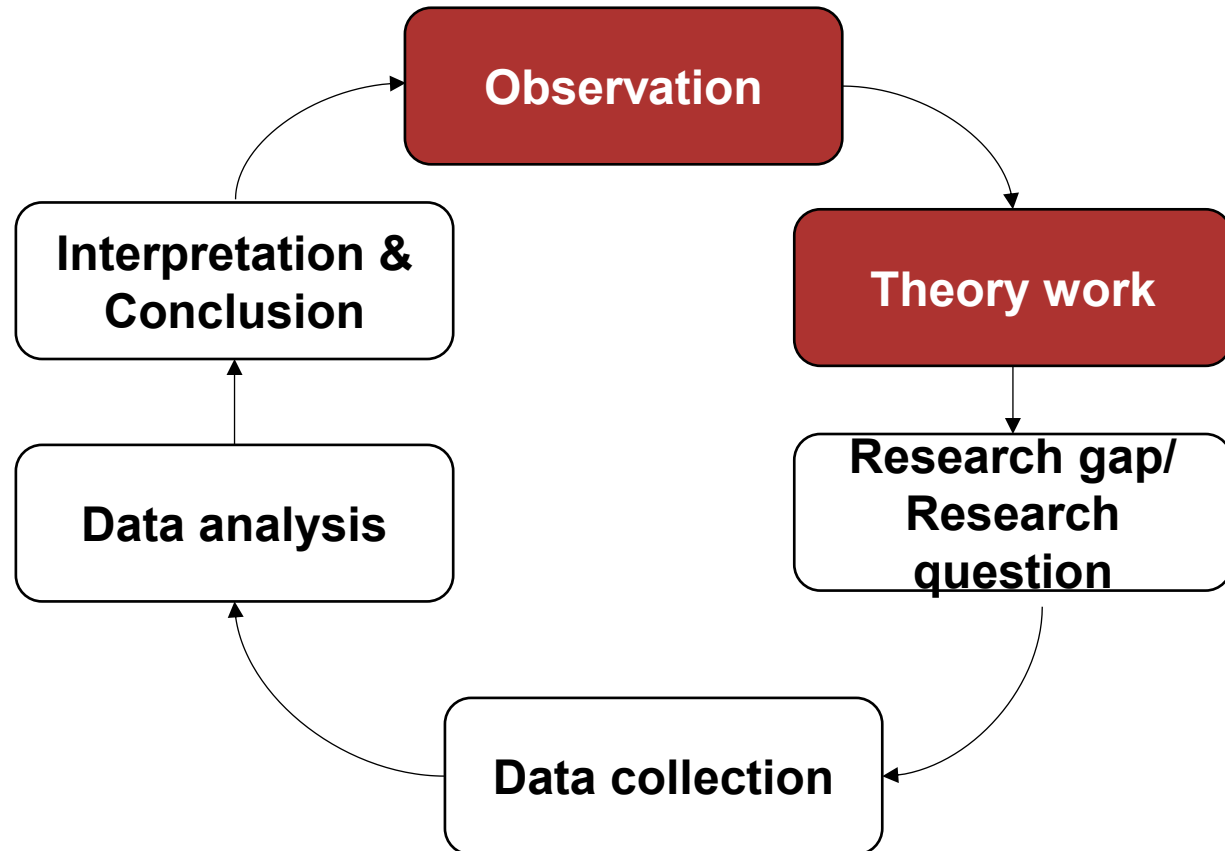
<https://lennartnacke.com/how-to-write-a-chi-paper/>

Reflection



How did it go?
What went good/
needs refinement?
What was most
challenging?

Goal for today:



 Templates

Any questions so far?



Theory work

As a reminder:

- A scientific theory consists of abstract **constructs**, more concrete **concepts** and variables – aka how the concepts/ constructs are “operationalized” and become measurable **What are the core constructs & concepts in your projects? (5 -7 min)**

Back to plans



Remove image

TITLE *

#RS 6 How to write a theory section and introduction to the report template

DESCRIPTION

- Hands-on-session how to write a theory section

RESOURCES AND ACTIVITIES

- The ACM Template (e.g., for CHI 2026 and your assignments) 8/36
- Template_assignment_1.docx 12/36
- What are the core constructs or concepts in your project? 0/36
- Digital Whiteboard for todays session 0/36

Add

Whiteboard

Whiteboard

Lena Frischlich LF

Teilen

Text hinzufügen

Team Anne: AI discussions on Twitter

Team Hannah: AI chat-bots as voting advisors

Text hinzufügen

What are the core concepts/ constructs in your project?

Team Lynga: AI-generated news

Text hinzufügen

Team Lena: Food cues & discussions on YouTube

Text hinzufügen

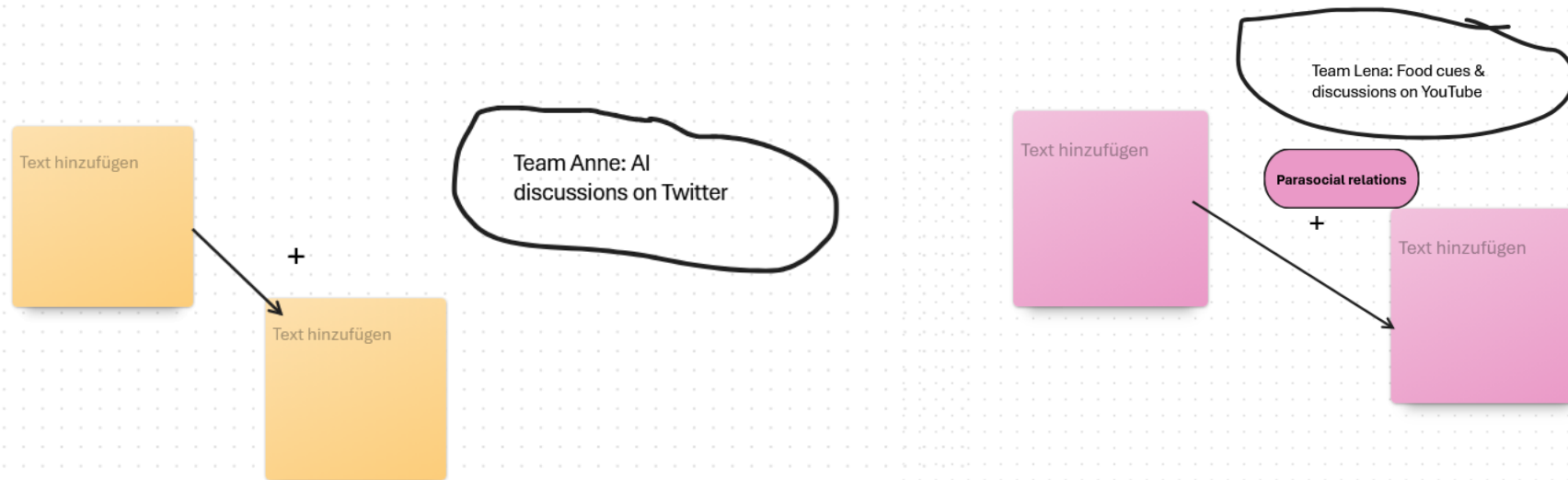
Team Curd: Alternative news coverage of contested topics

Text hinzufügen

65 %

A theory links variables/ concepts to each other

- How are your concepts linked? (5-7 min)
- Add arrows and qualifiers (+ / - / ?) to your postit's
- Why do you believe that? Use “labels” to add theories or “logical conclusion” to your links



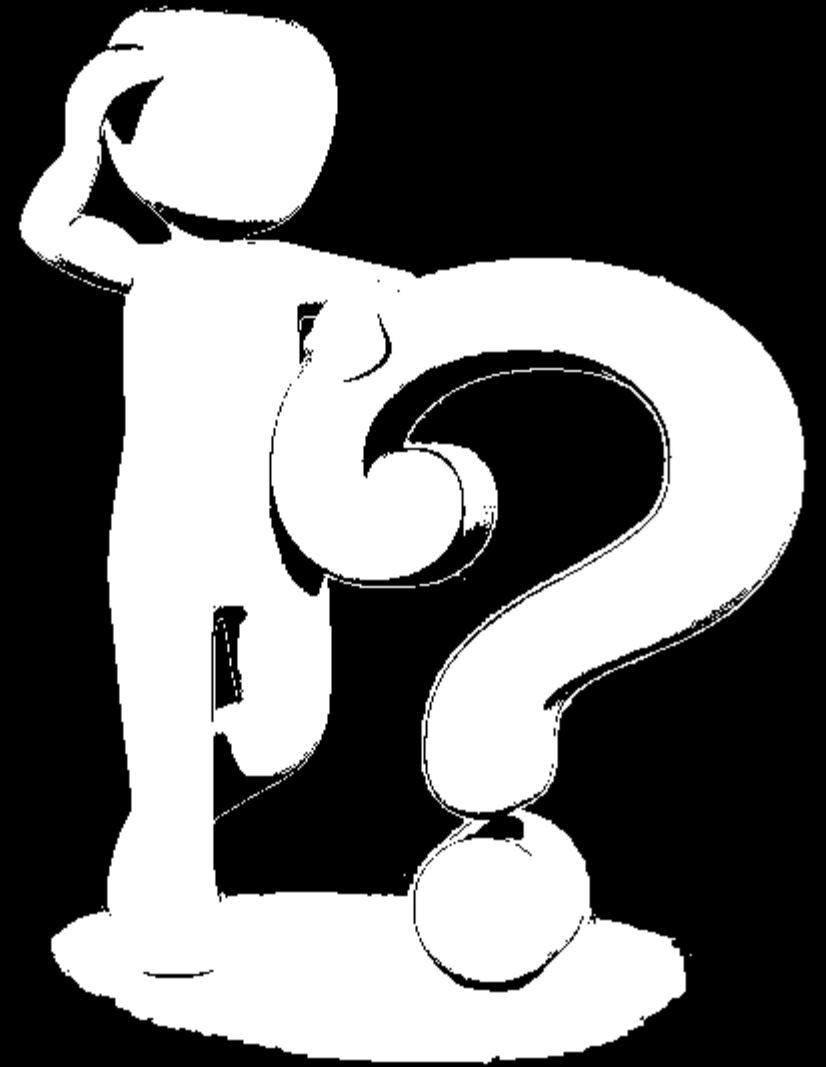
Remember: types of theoretical statements

- a. **Hypothesis:** a testable statement about the relationship between two or more concepts (variables) – can relate to differences $A > B$ or relations $A+ \rightarrow B+ \Rightarrow$ Your +/-
- b. **Research question:** an open question about the relationship between two or more concepts $\Rightarrow$ Cannot be answered by “yes” or “no” (else it is likely a hypothesis ;-))
 - Can be distinguished from
 - **Propositions** are merely descriptive and provide information about only one variable at a time – less useful than hypothesis (no arrows or qualifiers attached)
 - **Assumptions:** A theoretical statement that is taken for granted (or untestable). May describe the relationship between variables (similar to a hypothesis), or the usual value of one (as in propositions).

Go back to the whiteboard (5 -7 min)

- What is a research question link (?)
- What is a hypothesis (+/-)
- Are there propositions involved? (i.e., post-it's without arrows/ descriptions focussing only on one variable?) - can you qualify what you expect from them? (+/-/ ?)
- Are there assumptions involved that cannot be tested/ will not be tested? Mark them with a 

How was it?
What were challenges?
What went well?
What is missing?



Time for a break



How to write & present a theory section

How does the scientific circle inform your theory pitch?

Convincing presentations implement a “story arch”

Theory work is the first part of this “story”

(1) Context

(= introduction + theory to set the scene & grab attention)

(2) Conflict (= identify the problem)

Afterwards:

(3) Present the **solution**

(= your methodological approach + results/ key insights)

(4) **Resolution** (conclusion):

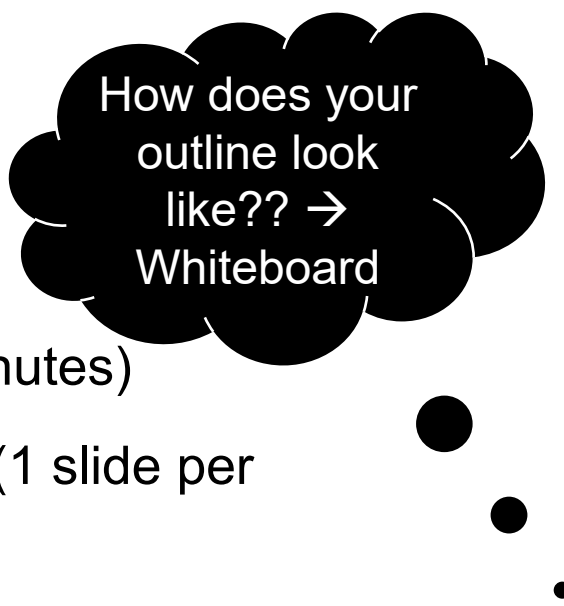
reinforce key message + implications

You have 10-12 minutes in total

Introduction (~ 1 slide/ 1 minute)

Theoretical background (~ 8 minutes)

1. Central construct/ concepts (1 slide per construct)
2. Central theoretical links (1 slide per theory)
3. What has not been studied so far, what is the problem that this study will solve? (~ 2 minutes, end with the research questions/ hypotheses)



How does your outline look like?? → Whiteboard

How does the scientific circle inform your theoretical argument?

The arguments follow the same structure

Introduction (~ 3 paragraphs)

- a) Broad phenomenon/ observation
- b) Central construct/ concept & central theoretical links
- c) Central contribution of the current study

Theoretical background (with subsections)

- 1. Central construct/ concepts
- 2. Central theoretical links
- 3. Central problem: What has not been studied so far, what is the problem that this study will solve?

Also reflected in the template

Submission Template for Assignment 1 Theory and Research Questions¶

This is Short Title of the paper, used in page headers¶

This is the subtitle of the paper, this document both explains and embodies the submission format for authors using Word¶

First Author's Name¶¶

First author's affiliation, an Institution with a very long name, xxxx@student.sdu.dk¶

Second Author's Name¶¶

Second author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

Third Author's Name¶¶

Third author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

Fourth Author's Name¶¶

Fourth author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

Fifth Author's Name¶¶

Fifth author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

Sixth Author's Name¶¶

Sixth author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

Seventh Author's Name¶¶

Sixth author's affiliation, possibly the same institution, xxxx@student.sdu.dk¶

¶

Although there is no distinctive header, this is the abstract. Write here up to 100 words to introduce your project. Use the following logic: First or first two sentences(s): General introduction to the phenomenon/ problem, written such that everyone can understand it. Next one or two sentences: Describe the theoretical background of your project so that people familiar with the topic can understand the specific research gap that you seek to close here. Use formulations such as "prior work has done abc" and "while highly informative, this work did not xyz". Describe which overarching research question the project seeks to answer or which overarching hypothesis it seeks to test. Examples include: "The following study thus seeks to understand/ test/ explore/ examine..."¶

KEYWORDS/ CORE CONCEPTS • Insert your first CCS term here • Insert your second CCS term here • Insert your third CCS term here¶

Additional Keywords and Phrases: Insert comma delimited author-supplied keyword list, Keyword number 2, Keyword number 3, Keyword number 4¶

¶

Word count: Aim for 1000- 4000 words (excluding references, tables, figures, and appendices)¶

¶

* Place the footnote text for the author (if applicable) here. ¶

1 → INTRODUCTION¶

Give an initial introduction into the topic/ the phenomenon that you want to study. Use the first paragraph to catch the readers' attention and formulate it in a way that lay audiences can understand what you are studying. A great tip is to start with a statement sentence and then use three or four sentences to justify your statement. Use the last sentence of the paragraph to conclude your statement and formulate a bridge to the next paragraph.¶

Use one paragraph per statement or argument. For example, your first paragraph could introduce in the context or phenomenon, while the second one narrows down what about this phenomenon on context is research worthy — what is missing from prior research?¶

Probably you need more than one paragraph to describe the research gap. For example, one paragraph could describe the central theory or concept you are working with, while the second one describes the central research on the context in which you are working (e.g., because your theory has not been tested on social media or in Danish language etc. pp).¶

The final¶ paragraph of your submission should present your research questions or hypotheses: Based on the previous paragraphs: What is missing from literature? Which research gap(s) or problem(s) did you identify (You can check out the following paper from CHI2025 as an example <https://dl.acm.org/doi/10.1145/3706598.3713829>). Remember: A research question should be formulated openly (not to be answered by yes or no, e.g., how is A related to B) while a hypothesis formulates a prediction (e.g., more A leads to more B).¶

2 → BACKGROUND¶

The background section fleshes the arguments that you made in the theory section out — try to use the same order of arguments as you have used in the introduction, that helps your readers to follow your train of thoughts and increases the rigorousness of your text. In the background section, you can use subheadings. Some typical subheadings are presented below — but you can choose headings that reflect the content of the respective section best.¶

2.1 → Core constructs and definitions¶

What are the core constructs and definitions that you are using in this manuscript? Give the reader a working definition. This enables future scholars to build on your work and others to know what exactly you are working on.¶

Remember that validity (= measures what it should measure), reliability (= measures always the same, independent of the researcher, setting etc.) and objectivity (= the measure is independent of the one who administers it) are core quality criteria in research. To judge them clear definitions of the concepts you are working with are mandatory.¶

Use one paragraph for each core concept that needs definition (aka is not "common knowledge" in the sense of your neighbor, mother, cranky uncle all have the same understanding of it).¶

2.2 → Theoretical background¶

What is the core theory that you use to link the concepts that you are working with to each other. What is the background for your predictions and which lenses do you use to answer your research questions?¶

Think about this section as a funnel: Start with the overarching theoretical framework and then narrow it down to the specific context that you are studying. Continue to use one paragraph per argument (e.g., paragraph 1 Theory of All Things (TAT), paragraph 2 How the TAT has been studied in computer-mediated communication, paragraph 3 prior work on the TAT in the context of AI agents.¶

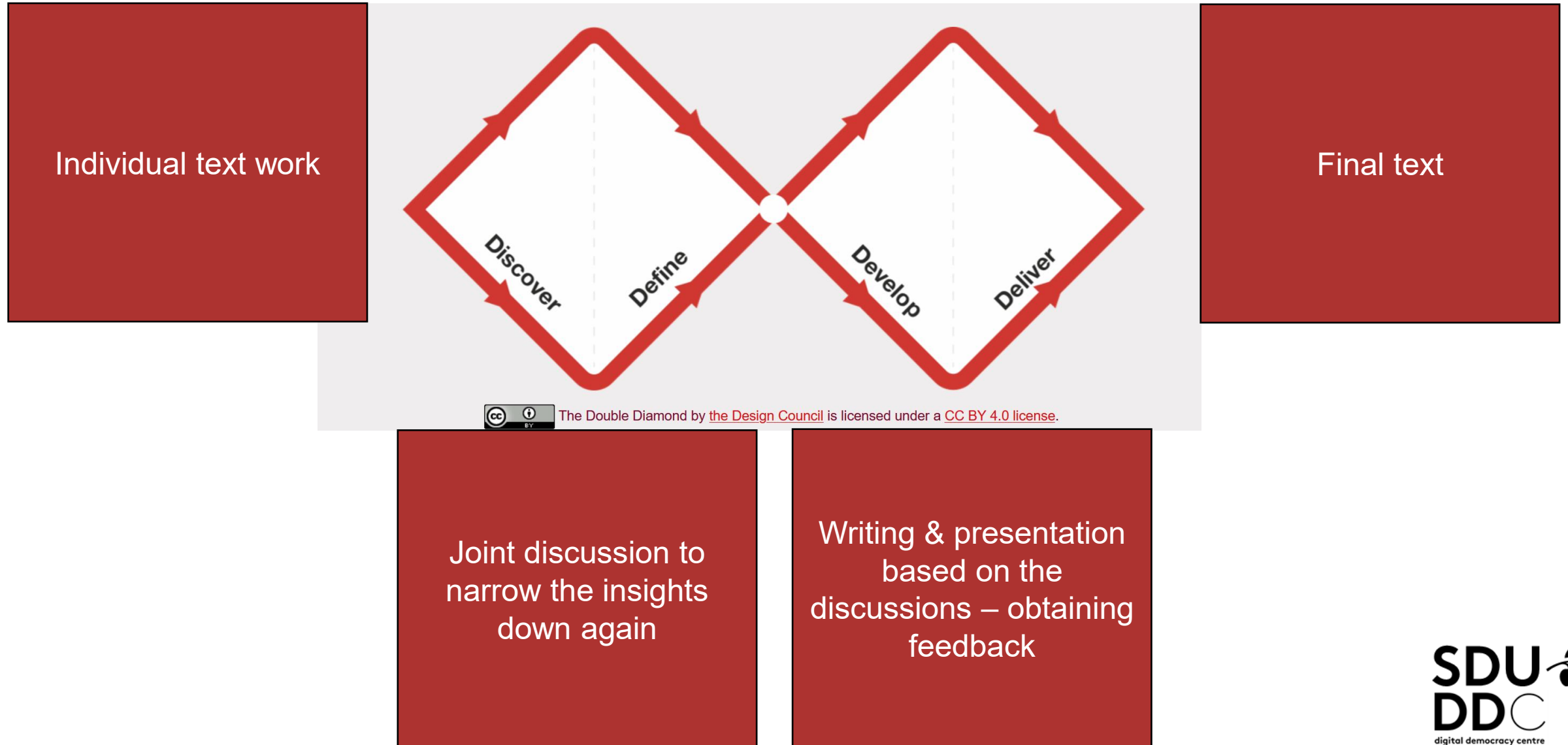
¶

¶ In a "real" ACM paper you would also write two more paragraphs about the methods you used to answer your research questions and the core results. This is NOT part of this first assignment¶

How to work with templates

- Templates enable the adaptation of the text to corporate identity or scientific standards
- Here we use the CHI template as this is one central standard in your field
 - Can be used for formatting in Overleaf
 - Can be opened in google docs
 - Can be used as a word template
- **Core aspects of the template**
 - Aim for 1000- (max!!!) 4000 words (excluding references and appendices such as the AI-statement) - plus/ minus 10% is fine, this is more to orient yourself
 - Use the power of the group: Create a joint “skeleton outline” (→ use your presentation) and distribute tasks for writing
 - Topic experts per section
 - “Product owner” per assignment to ensure consistency
 - With 7 people in the group that means ~ max. 500 words per section plus 500 for the introduction and the “problem” summary
- **Citation of scientific references**
- **NOT: Processor/ if used at all/ page layout**

Remember the double diamond?



From skeleton outline to actual text – the writing process

Introduction (~ 3 paragraphs) =>

Write last!

- a) Broad phenomenon/ observation
- b) Central construct/ concept & central theoretical links
- c) Central contribution of the current study

Theoretical background (with subsections)

1. Central construct/ concepts

- Definition concept 1
- Definition concept 2

2. Central theoretical links

- Broadest theoretical framework (e.g., Theory of Behaviour)
- Prior research on Theory + Phenomenon (e.g., Behaviour on Social media)
- Prior research on Context (e.g., Social Media in Denmark)

3. **Central problem:** What has not been studied so far, what is the problem that this study will solve? → All together (e.g., Behaviour on social media in Denmark)

From headlines to paragraphs

Example: 2) Central theoretical links

Broadest theoretical framework (e.g., Theory of Behaviour)

- Introduction of theory
- What do we know about the theory (has it been tested/ revised?)
- Current state of the theory (e.g., mounting evidence in line with the theory versus conflicting evidence?)

Prior research on Theory + Phenomenon (e.g., Behaviour on Social media)

- Focus on the core results – start with reading the abstract and discussion
→ That is what you need to describe the study & the discussion typically mentions limitations/ need for future research
- Use the theory section to identify further relevant literature/ use Google Scholar to see who has cited it recently
- Start with a statement – then use the literature as your witness
- E.g., Behaviour on social media is different. Using an experiment/ interviews, Study 1 showed...

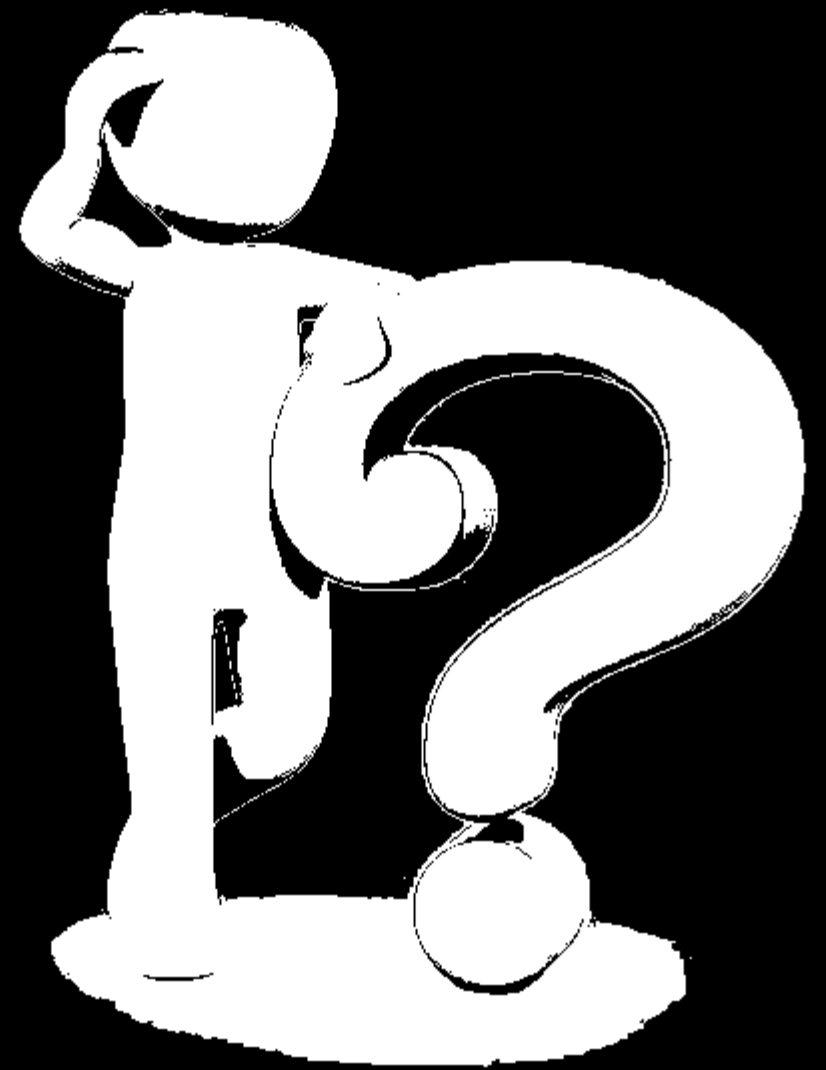
How can AI support you?

Where can AI support the process? What do you think? A systematic assessment

Stage	Potential(s)	Risk(s)	Conclusion
Description of the phenomenon			
Creation of the skeleton outline			
Definition of core constructs			
Summarizing of core theorie(s) and scientific articles			
Identifying new texts			
Proof-reading			
Formatting the final document			

**So: Get to work – you can
do this!**

Any more questions?



**AI 507 –
Research seminar
AI & Society**

Have a wonderful rest of the day! Remember – the next step is meeting your supervisor – and we meet for the first presentation in 2 weeks – THURSDAYS LECTURE IS A VIDEO!